

BUSINESS REQUIREMENTS DOCUMENT

Automated Banking Analytics Dashboard

SQL Server • Microsoft Fabric • Power BI — End-to-End Banking Intelligence Solution

Document Type	Business Requirements Document (BRD)
Project Name	Automated Banking Analytics Dashboard
Domain	Retail Banking — Transactions, Accounts & Customer Analytics
Prepared By	Senior Business Analyst
Prepared For	Banking Operations, Retail Leadership & Data/BI Team
Version	1.0
Status	Final

1. Executive Summary

The bank currently records millions of customer transactions across six channels (ATM, Mobile Banking, UPI, Internet Banking, Branch, and POS) but lacks a unified, automated view of transaction health, branch performance, and customer behaviour. Business teams rely on manual extracts and static spreadsheets, which delay decisions on fraud/failure hotspots, underperforming branches, and cross-sell opportunities.

This project delivers an Automated Banking Analytics Dashboard that connects directly to the bank's SQL Server transaction system, cleans and models the data through Microsoft Fabric, and surfaces it through a live, self-refreshing Power BI report. In parallel, a set of 10 structured SQL business-analysis queries convert raw transaction data into ranked, actionable findings (e.g., which branches need intervention, which channels are failing, which customers are ready for cross-sell).

The solution removes manual reporting effort, refreshes automatically every day, and gives Operations, Branch Management, and Customer Strategy teams a single source of truth for day-to-day decisions.

2. Business Objectives

- Provide a real-time, automated view of transaction volume, value, and success rate across the bank.
- Identify underperforming branches and channels so operations can intervene early.
- Segment customers by value, income, and behaviour to support targeted cross-sell campaigns.
- Reduce manual MIS/reporting effort by automating data ingestion, transformation, and refresh.
- Convert raw transactional data into ranked, decision-ready business recommendations, not just charts.
- Establish a scalable data model (star schema) that can extend to new products and channels.

3. Stakeholders

Stakeholder	Role	Interest in Project
Retail Banking Leadership	Business Sponsor	Strategic view of transaction & customer performance
Branch Operations Team	Primary User	Identify branches/channels needing intervention
Customer Strategy / Marketing	Primary User	Cross-sell targeting, segment-based campaigns
Risk & Fraud Team	Secondary User	Monitor failure rate concentration by channel/segment
Data & BI Team	Build & Maintain	Own Fabric pipeline, semantic model, and dashboard
IT / Infrastructure	Support	SQL Server hosting, On-Premises Data Gateway, security

4. Project Scope

4.1 In Scope

- Ingestion of 5 core tables from SQL Server: FactTransaction, DimAccount, DimBranch, DimCustomer, DimDate.
- Data cleaning and calculated column creation (Age Band, Income Band, Interest Rate Band) via Power Query in Dataflow Gen2.
- Star-schema semantic model with a centralized measures table for all DAX KPIs.
- Two-page interactive Power BI report: Executive Overview and Customer Insights.
- 10 SQL-based business analysis queries covering failures, branch performance, segmentation, and cross-sell.
- Fabric Data Pipeline for automated daily refresh (Dataflow Gen2 → Semantic Model).
- Custom brand-aligned visual theme applied consistently across both report pages.

4.2 Out of Scope

- Real-time (sub-daily) streaming refresh — current scope is a daily scheduled refresh.
- Predictive modelling / machine learning (e.g., churn prediction, fraud scoring models).
- Mobile app development — delivery is via Power BI Service / embedded report only.
- Write-back or transactional actions from the dashboard (view-only analytics).

5. Current State & Problem Statement

Transaction, account, branch, and customer data reside in a normalized SQL Server database with no automated reporting layer. Business questions such as “which channel has the highest failure rate” or “which customers hold only one account” require manual SQL extraction and spreadsheet analysis, typically taking hours to days to answer. There is no shared, governed definition of KPIs like Success Rate or Accounts per Customer, causing inconsistent numbers across teams.

This project addresses these gaps by building a governed semantic model with centrally-defined measures, a scheduled refresh pipeline, and a curated set of business-analysis queries that translate directly into operational actions.

6. Data Architecture & Technical Approach

6.1 Architecture Overview

- Source System: SQL Server database (Banking_Domain) hosted on an on-premises server.
- Connectivity: On-Premises Data Gateway links SQL Server to Microsoft Fabric.
- Ingestion & Cleaning: Dataflow Gen2 (Power Query / M) extracts, cleans, and enriches the data.
- Storage: Cleaned Delta tables land in the Banking Lakehouse under the dbo schema.
- Modelling: A star-schema semantic model is built on the Fabric SQL Analytics Endpoint.
- Business Analysis: 10 structured SQL Server queries run against the semantic model layer to answer ranked business questions (see Section 9).
- Visualization: Power BI report connects via Live Connection to the semantic model.
- Automation: A Fabric Data Pipeline sequences Dataflow Gen2 → Semantic Model refresh, scheduled daily at 10:00 PM.

6.2 Source Data Model

Table	Grain / Purpose	Key Attributes
FactTransaction	One row per transaction (fact)	TransactionKey, DateKey, CustomerKey, AccountKey, BranchKey, Channel, Amount, Status
DimAccount	One row per account	AccountKey, AccountType, AccountStatus, OpenDate, CreditLimit, InterestRate
DimBranch	One row per branch	BranchKey, BranchName, City, State, Region, BranchType
DimCustomer	One row per customer	CustomerKey, DOB, Occupation, CustomerSegment, AnnualIncome, CustomerSince
DimDate	One row per calendar date	DateKey, Date, Month, MonthName, Quarter, Year, IsWeekend

6.3 Data Cleaning & Calculated Columns (Dataflow Gen2 / Power Query)

- Surrogate keys (Account/Customer/Branch/Date Key) converted from text to Whole Number.
- Date fields (Open Date, Date of Birth, Customer Since) standardized to strict Date type.
- Age Band derived from Date of Birth: 18–25, 26–35, 36–45, 46–55, 56–65, 65+.
- Income Band derived from Annual Income: <3L, 3–8L, 8–15L, 15L+.
- Interest Rate Band derived from Account Interest Rate: 0–4%, 4–7%, 7–10%, 10–13%, 13%+.
- All transformed tables are loaded into the Lakehouse in Delta format for query performance.

7. Semantic Model & KPI Definitions

All DAX measures are centralized in a dedicated Measures table within the semantic model (rather than in Power BI Desktop) to keep the model lightweight and ensure consistent, governed KPI definitions across every report and future consumer.

7.1 Executive Overview KPIs

KPI	Business Definition	DAX Logic
Total Transaction Amount	Overall value processed	SUM(TransactionAmount)
Transaction Count	Volume of transactions	COUNTROWS(FactTransaction)
Average Transaction Value	Typical ticket size	Total Amount ÷ Count
Success Rate	% transactions completed successfully	Successful Count ÷ Total Count

7.2 Customer Insights KPIs

KPI	Business Definition	DAX Logic
Total Customers	Unique customer base size	DISTINCTCOUNT(CustomerID)
Average Annual Income	Customer earning profile	AVERAGE(AnnualIncome)
Average Interest Rate	Average rate across accounts	AVERAGE(InterestRate)
Accounts per Customer	Product depth / cross-sell indicator	Distinct Accounts ÷ Distinct Customers

Dynamic Field Parameters allow business users to toggle chart metrics without needing a separate visual for each KPI, on both the Executive Overview page (4 transaction metrics) and the Customer Insights page (4 customer metrics).

8. Dashboard Design & Visualization Requirements

8.1 Page 1 — Executive Overview

- KPI header cards: Total Transaction Amount, Transaction Count, Average Transaction Value, Success Rate.
- Monthly trend area chart with dynamic title bound to the selected KPI.
- Channel distribution donut chart (ATM, Mobile Banking, UPI, Internet Banking, POS, Branch) with center total.
- Regional performance bar chart (North, Central, East, West, South), sorted by value.
- Interactivity: Year selector, City/Region slicers, Key Metrics selector, Clear All Filters.

8.2 Page 2 — Customer Insights

- KPI header cards: Total Customers, Average Annual Income, Average Interest Rate, Accounts per Customer.
- Customer Segment donut (Mass, Affluent, Premium) with center total.
- Occupation distribution column chart across 10 profession categories.

- Income Band and Age Band bar charts for demographic analysis.
- Shared City/Region filters and Key Metrics selector, consistent with Page 1.

8.3 Visual Design Standard

Both pages follow one consistent brand palette (deep purple / royal blue gradient family) applied to KPI accents, chart series, selected-state controls, and donut center callouts, ensuring the report reads as a single cohesive product rather than two disconnected pages.

9. Business Analysis Requirements (SQL Layer)

Beyond the visual dashboard, the project defines 10 structured business questions, each resolved through a SQL Server query against the Banking_Domain model. Every question follows the same chain: Business Problem → SQL Analysis → Finding → Business Impact → Recommendation.

#	Business Question	SQL Concept
1	Which customer segments generate the most transaction value?	GROUP BY + JOIN
2	Which branches perform well or need attention?	CTE + CASE
3	Which channels are most successful?	Aggregation
4	Where are transaction failures concentrated?	Conditional Aggregation
5	Which account types are most valuable?	JOIN + Aggregation
6	Which customers have cross-sell potential?	CTE + CASE
7	Does income relate to banking activity?	GROUP BY
8	How is transaction performance changing monthly?	Window Function (LAG)
9	Which customer groups have high failure rates?	Multi-table JOIN
10	Where should management focus operational improvement?	CTE + KPI Logic

Outputs from these queries feed directly into operational reviews — e.g., Question 10 produces a ranked priority list combining revenue-at-risk, failure rate, and transaction volume, giving leadership a single actionable shortlist each quarter.

10. Automation, Scheduling & Refresh Requirements

- A Fabric Data Pipeline sequences the Dataflow Gen2 refresh followed by the Semantic Model refresh.
- Scheduled to run automatically once daily at 10:00 PM, requiring no manual trigger.
- On failure, the pipeline should alert the Data & BI team for same-day remediation (operational requirement).
- Live Connection ensures the Power BI report always reflects the latest refreshed semantic model, with no separate report-level refresh needed.

11. Assumptions & Constraints

11.1 Assumptions

- Source SQL Server data is reasonably clean at the row level; cleansing focuses on type conversion and derived bands, not deduplication.
- On-Premises Data Gateway remains available and licensed for the life of the project.
- Daily refresh cadence is sufficient for current business decisions (no intraday requirement today).

11.2 Constraints

- Live Connection semantic models limit report-level calculated columns; new columns must be added upstream in Dataflow Gen2 or the model.
- Fabric capacity and gateway throughput bound how much historical data can be reprocessed in a single refresh window.

12. Success Criteria & Expected Business Benefits

- Manual MIS/report preparation time reduced through fully automated daily refresh.
- Consistent, single-source KPI definitions eliminate cross-team reporting discrepancies.
- Branch and channel issues (failures, underperformance) surfaced within one refresh cycle instead of ad-hoc requests.
- Cross-sell target lists (single-account customers, low-product income bands) available on demand via SQL layer.
- A reusable star-schema foundation that can onboard new products/channels with minimal rework.

13. Sign-off

Role	Name	Signature / Date
Business Sponsor	_____	_____
Business Analyst	_____	_____
Data & BI Lead	_____	_____